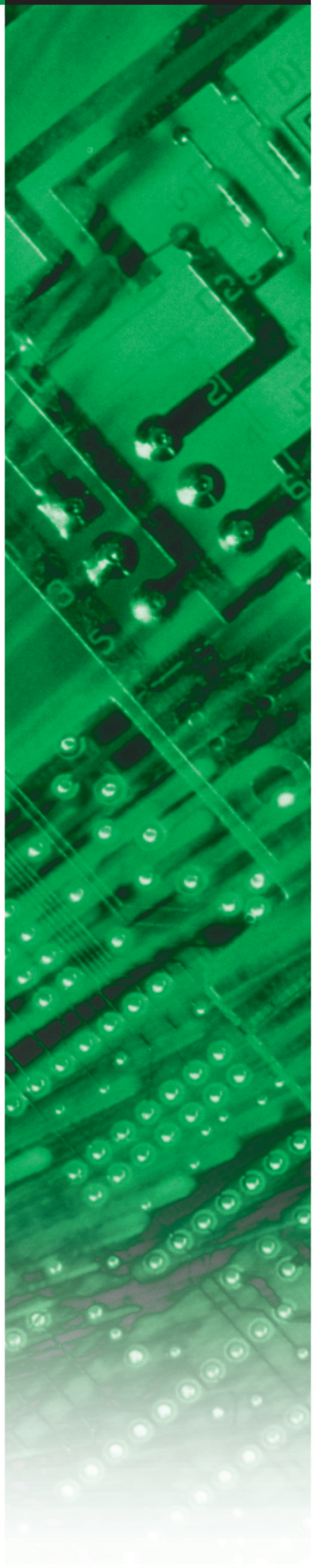


Rooftop Packaged Air Conditioners

Models: MRT 055 A/AR
MRT 060 A/AR
MRT 080 A/AR
MRT 100 A/AR
MRT 120 A/AR
MRT 150 A/AR
MRT 200 A/AR
MRT 250 A/AR
MRT 300 A/AR
MRT 360 A/AR
MRT 420 A/AR

M4RT 060 A/AR
M4RT 080 A/AR
M4RT 100 A/AR
M4RT 120 A/AR
M4RT 150 A/AR
M4RT 200 A/AR
M4RT 250 A/AR
M4RT 300 A/AR
M4RT 360 A/AR
M4RT 420 A/AR

McQuay® Air Conditioning

Engineered for flexibility and performance.™

Table of Contents

This manual supercedes MRT-2009

Note : Installation and maintenance are to be performed only by qualified personnel who are familiar with local codes and regulations, and experienced with this type of equipment.

Caution : Sharp edges and coil surfaces are a potential injury hazard. Avoid contact with them.

Warning : Moving machinery and electrical power hazard. May cause severe personnel injury or death. Disconnect and lock off power before servicing equipment.

CE

Nomenclature

Production spec variation

Type of ref. connection
X: Not Applicable

Electrical
F: 380-415V/3Ph/50Hz

M	4	RT	100	A	R	-	F	X	AA

Model Type
R: Heatpump
Blank: Cooling Only

Series
A: A series

Capacity
100: 100,000 Btu/h

Model Name
RT: Rooftop

Refrigerant
4: R407C
Blank: R22

Brand
M: McQuay

Product Line-up

M(4)RT Cooling Only

Model	Cooling Only	Nomenclature	Classification											
			SLM Controller	Seq. Controller	Capil lary Tube	Thermal Expansion Valve (TXV)	Normal Fin	Gold (NA549) Fin	Scroll Compressor	Air Filter	Down Flow	Side Flow	Convertible	Filter Drier
	060A	FXAB	X			X	X		X	X		X		X
	060A	FXAD	X		X		X	X	X	X		X		X
	080A	FXAA	X		X		X		X	X		X	X	X
	080A	FXAC	X		X		X	X	X	X		X	X	X
	100A	FXAA	X		X		X		X	X		X	X	X
	100A	FXAC	X		X		X	X	X	X		X	X	X
	120A	FXAB	X			X	X		X	X		X	X	X
	120A	FXAC	X			X	X	X	X	X		X	X	X
M4RT	150A	FXAA		X	X		X		X	X		X	X	X
	150A	FXAD		X	X		X	X	X	X		X	X	X
	200A	FXAA		X	X		X		X	X		X	X	X
	200A	FXAD		X	X		X	X	X	X		X	X	X
	250A	FXAA		X		X	X		X			X		X
	250A	FXAC		X		X	X	X	X			X		X
	300A	FXAA		X		X	X		X			X		X
	300A	FXAC		X		X	X	X	X			X		X
	360A	FXAC		X		X		X	X	X		X		X
	420A	FXAC		X		X		X	X	X		X		X
	055A	FXAI	X			X	X		X			X		X
	060A	FXAH	X			X		X	X	X		X		X
	060A	FXAI	X			X	X		X	X		X		X
	080A	FXAC	X		X		X		X	X		X		
	080A	FXAH	X		X			X	X	X		X		
	080A	FXAI	X			X	X		X	X		X	X	X
	080A	FXAN	X		X		X		X	X		X	X	
	100A	FXAC	X		X		X		X	X		X		
	100A	FXAH	X		X			X	X	X		X		
	100A	FXAI	X			X	X		X	X		X	X	X
	100A	FXAN	X		X		X		X	X		X	X	
	120A	FXAH	X			X		X	X	X		X	X	X
	120A	FXAI	X			X	X		X	X		X	X	X
	150A	FXAC		X	X		X		X	X		X		
	150A	FXAH		X	X			X	X	X		X		
MRT	150A	FXAI		X		X	X		X	X		X	X	X
	150A	FXAN		X	X		X		X	X		X	X	
	200A	FXAC		X	X		X		X	X		X		
	200A	FXAH		X	X			X	X	X		X		
	200A	FXAI		X		X	X		X	X		X	X	X
	200A	FXAN		X	X		X		X	X		X	X	
	250A	FXAA		X		X	X		X	X		X		X
	250A	FXAD		X		X		X	X	X	X			X
	250A	FXAH		X		X		X	X	X		X		X
	300A	FXAA		X		X	X		X	X		X		X
	300A	FXAD		X		X		X	X	X	X			X
	300A	FXAH		X		X		X	X	X		X		X
	360A	FXAA		X		X	X		X	X		X		X
	360A	FXAH		X		X		X	X	X		X		X
	420A	FXAA		X		X	X		X	X		X		X
	420A	FXAH		X		X		X	X	X		X		X

M(4)RT Heat Pump

Model	Heat Pump	Nomenclature	Classification												
			SLM Controller	Seq. Controller	Capillary Tube	Thermal Expansion Valve (TXV)	Normal Fin	Gold (NA549) Fin	Reciprocating Compressor	Scroll Compressor	Air Filter	Down Flow	Side Flow	Convertible	Filter Drier
M4RT	060AR	FXAC	X			X		X		X	X		X		X
	080AR	FXAC	X			X		X		X	X		X	X	X
	100AR	FXAC	X			X		X		X	X		X	X	X
	120AR	FXAC	X			X		X		X	X		X	X	X
	150AR	FXAC		X		X		X		X	X		X	X	X
	200AR	FXAC		X		X		X		X	X		X	X	X
	250AR	FXAC		X		X		X		X	X		X		X
	300AR	FXAC		X		X		X		X	X		X		X
	360AR	FXAC		X		X		X		X	X		X		X
	420AR	FXAC		X		X		X		X	X		X		X
MRT	055AR	FXAI	X			X	X			X			X		X
	060AR	FXAH	X			X		X		X	X		X		X
	060AR	FXAI	X			X	X			X	X		X		X
	080AR	FXAC	X		X		X		X		X		X		
	080AR	FXAH	X		X			X	X		X		X		
	080AR	FXAN	X		X		X		X		X		X	X	
	100AR	FXAC	X		X		X		X		X		X		
	100AR	FXAH	X		X			X	X		X		X		
	100AR	FXAN	X		X		X		X		X		X	X	
	120AR	FXAH	X			X		X		X	X		X	X	X
	120AR	FXAI	X			X	X			X	X		X	X	X
	150AR	FXAC		X	X		X		X		X		X		
	150AR	FXAH		X	X			X	X		X		X		
	150AR	FXAN		X	X		X		X		X		X	X	
	200AR	FXAC		X	X		X		X		X		X		
	200AR	FXAH		X	X			X	X		X		X		
	200AR	FXAN		X	X		X		X		X		X	X	
	250AR	FXAA		X		X	X			X	X		X		X
	250AR	FXAD		X		X		X		X	X	X			X
	250AR	FXAH		X		X		X		X	X		X		X
	300AR	FXAA		X		X	X			X	X		X		X
	300AR	FXAD		X		X		X		X	X	X			X
	300AR	FXAH		X		X		X		X	X		X		X
	360AR	FXAH		X		X		X		X	X		X		X
	420AR	FXAH		X		X		X		X	X		X		X

Features

High Efficiency
The McQuay rooftop unit is supplied with high efficiency and reliable scroll compressor (except MRT 080/100/150/200 AR).

Package Unit
Dubbed the “Plug and Play” equivalent in the air-conditioning industry, the single unit configuration allows hassle free installation. No additional piping work is required as both the indoor and outdoor sides are pre-connected. Refrigerant is factory pre-charged to ensure clean and efficient operation.

Flexibility Of Air Supply
McQuay rooftop unit uses a belt driven fan such that the air volume and static required can be adjusted according to the requirement. This flexibility allows for wider application.

Flat Top Design
McQuay rooftop unit’s flat top design allows for maximum utilization of warehouse and container space.

Electrical Control Capability
All series are equipped with our Electronic Remote Controller (SLM and Sequential LCD). The Sequential LCD features 7 days programmable timer, compressor run/error status, cool/heat/fan/dry/auto modes, etc, while the SLM features a 15 hour delay timer, cool/heat/fan/dry/auto modes, etc.
Part loading of system capacity is now possible with the use of our Sequential PAC (Package Air-Condition) control for multiple compressor McQuay rooftop units (M(4)RT150/200/250/300/360/420 A/AR).

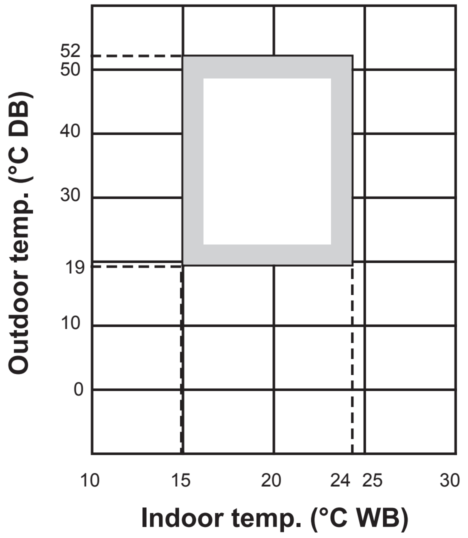
Application Information

Operating Range

Ensure the operating temperature is in allowable range.

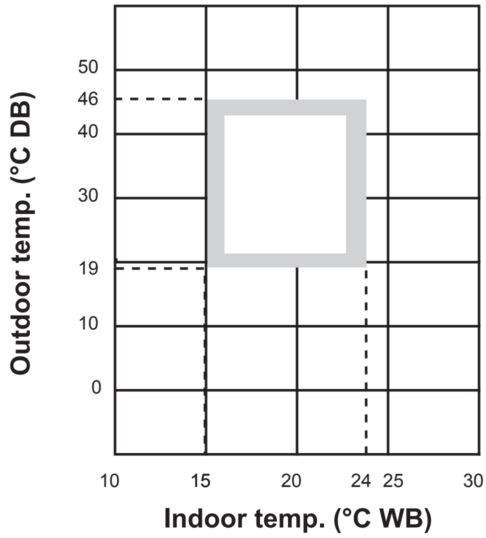
Cooling (R22)

Cooling Only Unit



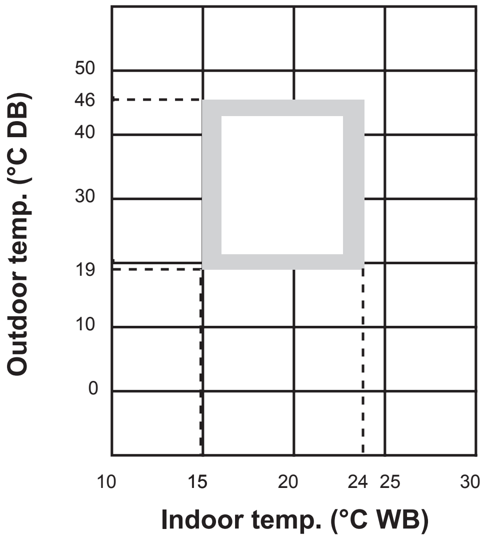
Cooling (R22)

Cooling Mode For Heat Pump Model



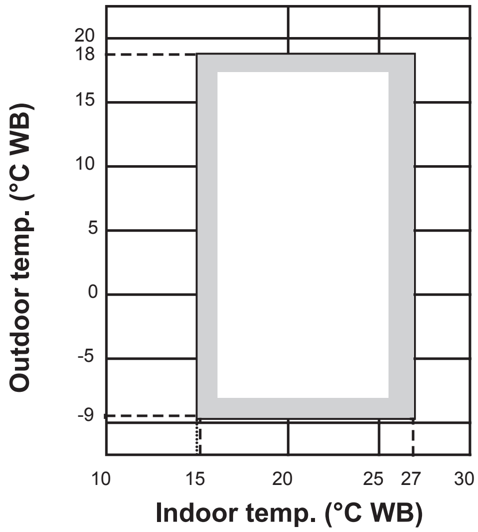
Cooling (R407C)

Cooling Only Unit & Cooling Mode For Heat Pump Model



Heat pump

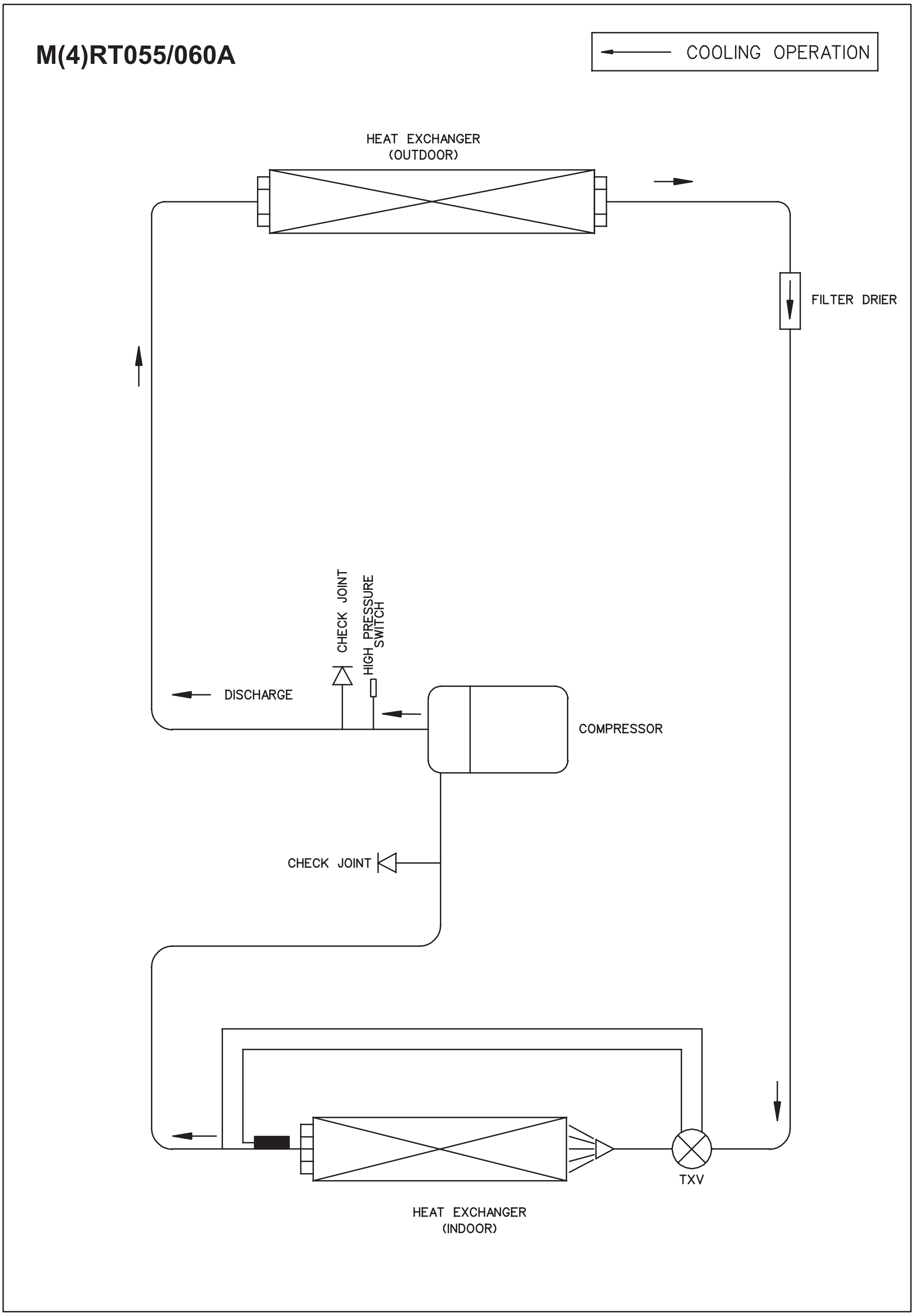
Heat Pump Unit Only



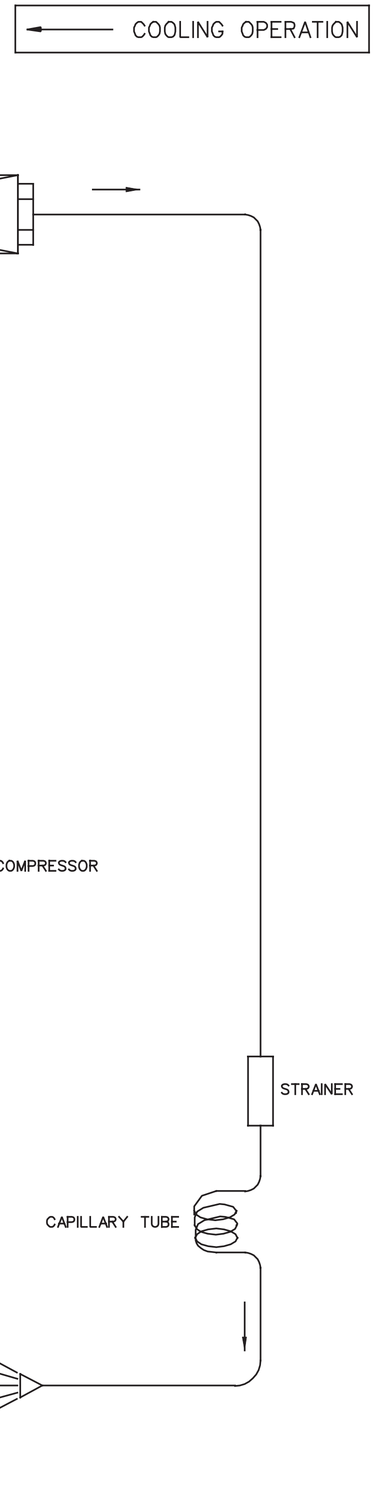
⚠ **Caution :**

The use of your air conditioner outside the range of working temperature and humidity can result in serious failure.

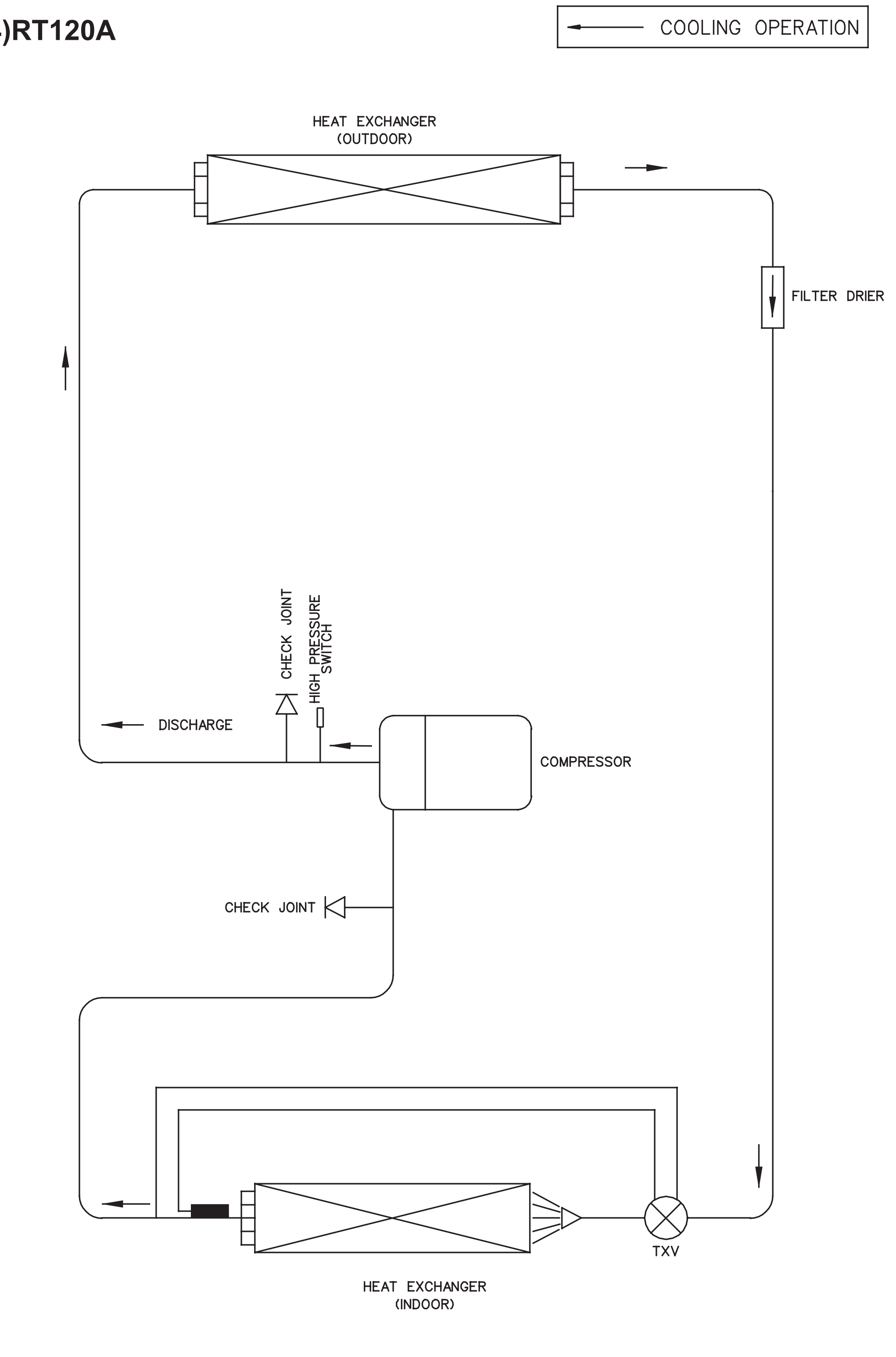
Refrigerant Circuit Diagrams



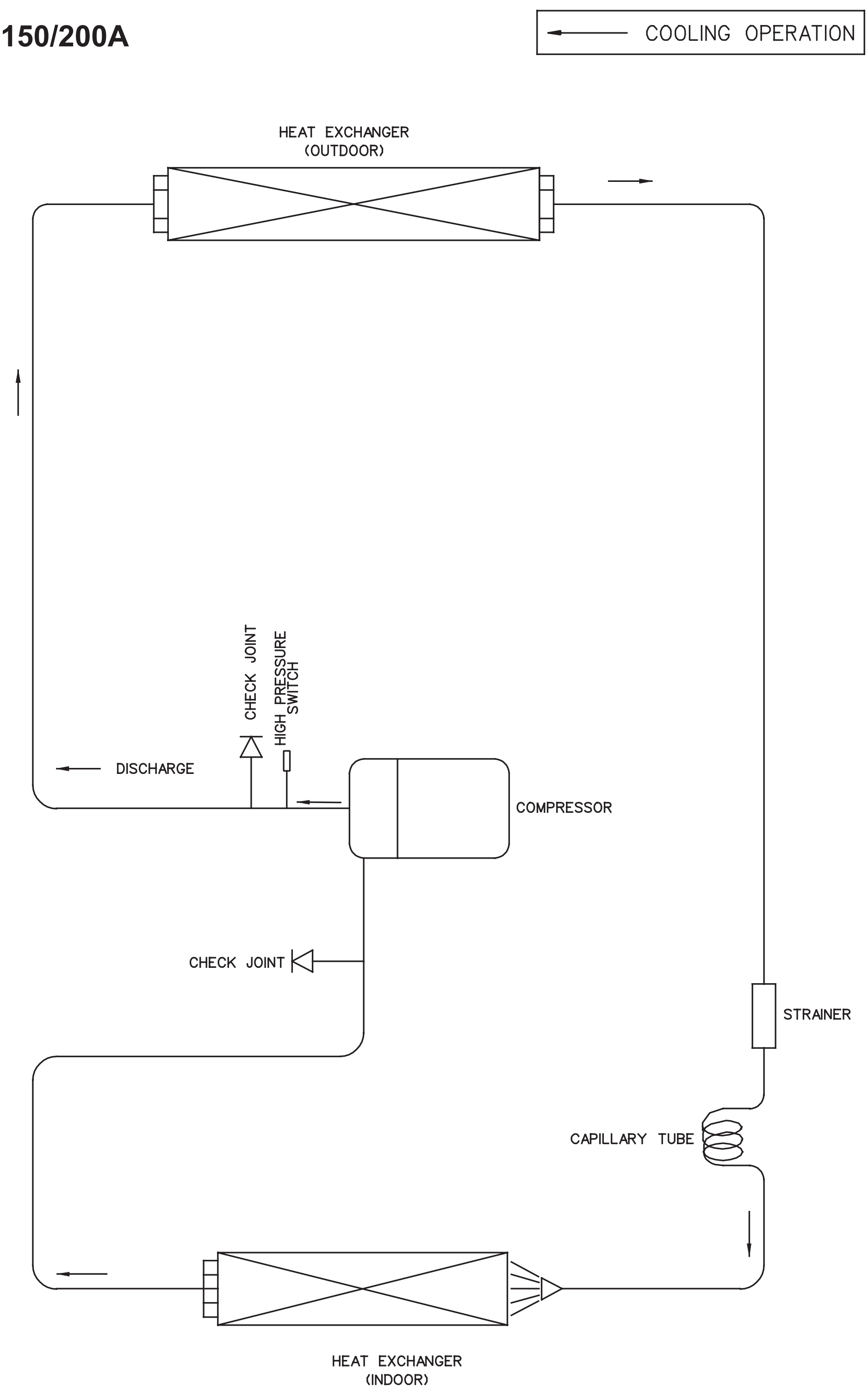
M(4)RT080/100A



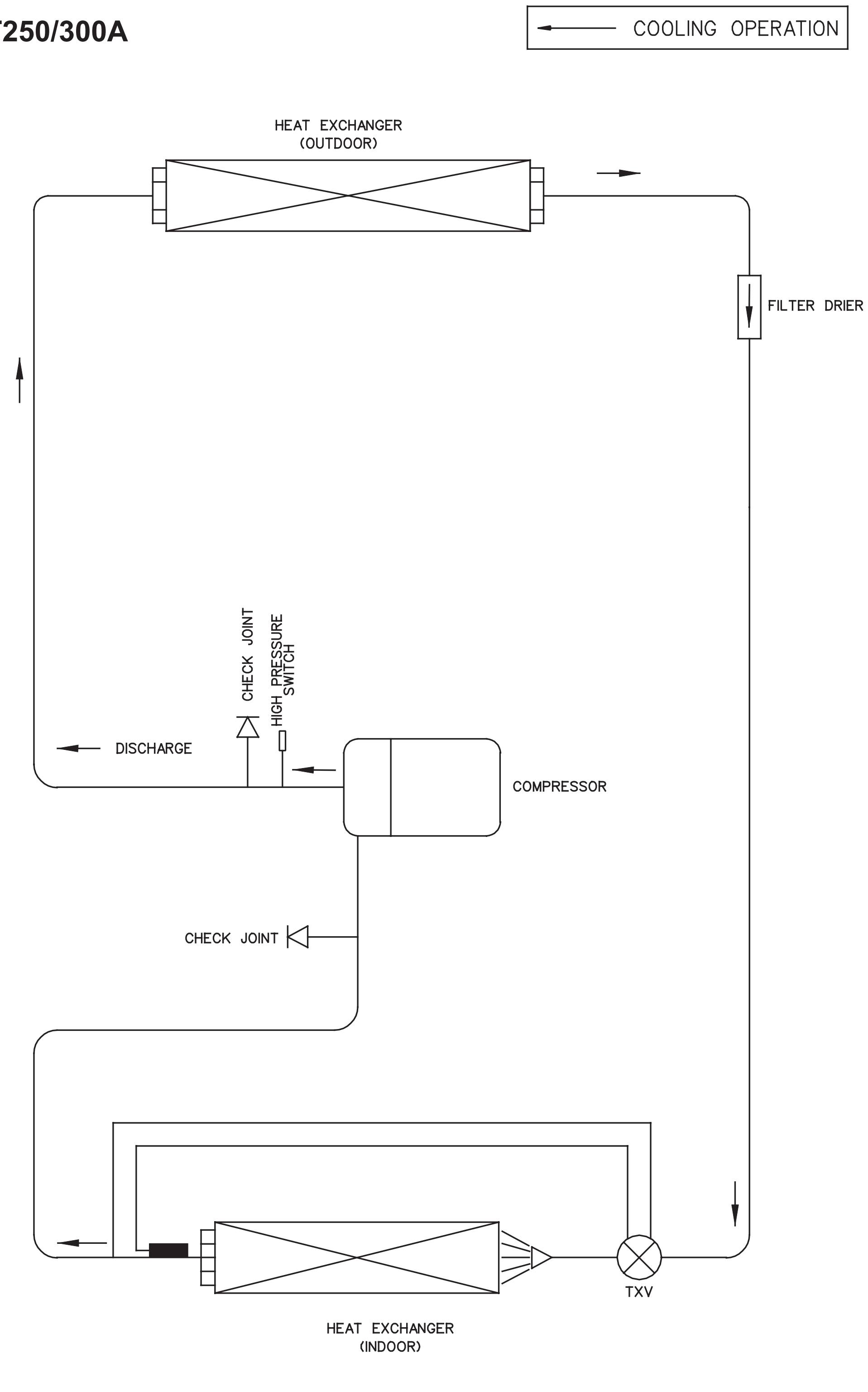
M(4)RT120A


COOLING OPERATION
HEAT EXCHANGER
(OUTDOOR)
FILTER DRIER
CHECK JOINT
HIGH PRESSURE
SWITCH
DISCHARGE
COMPRESSOR
CHECK JOINT
TXV
HEAT EXCHANGER
(INDOOR)

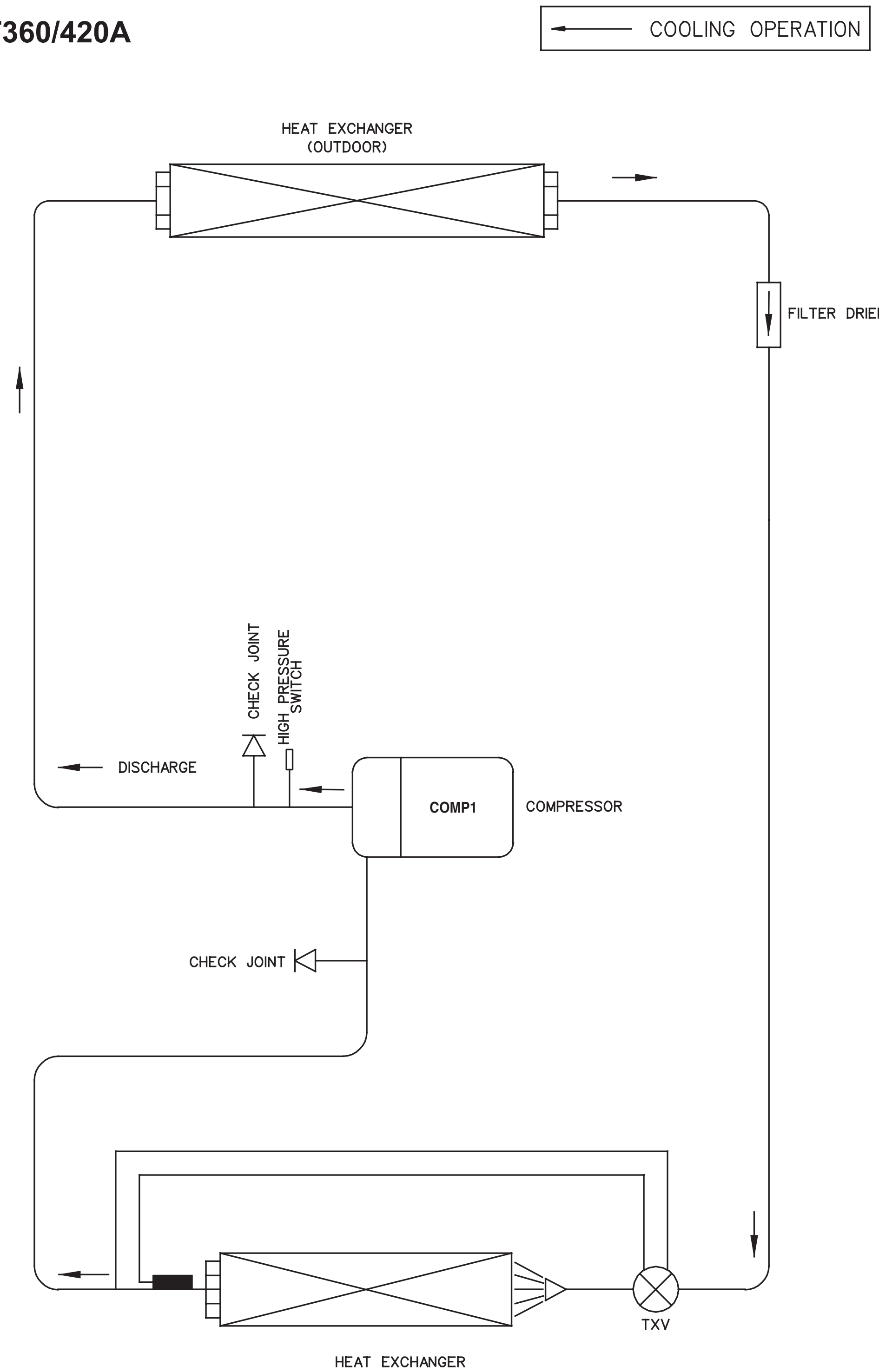
M(4)RT150/200A



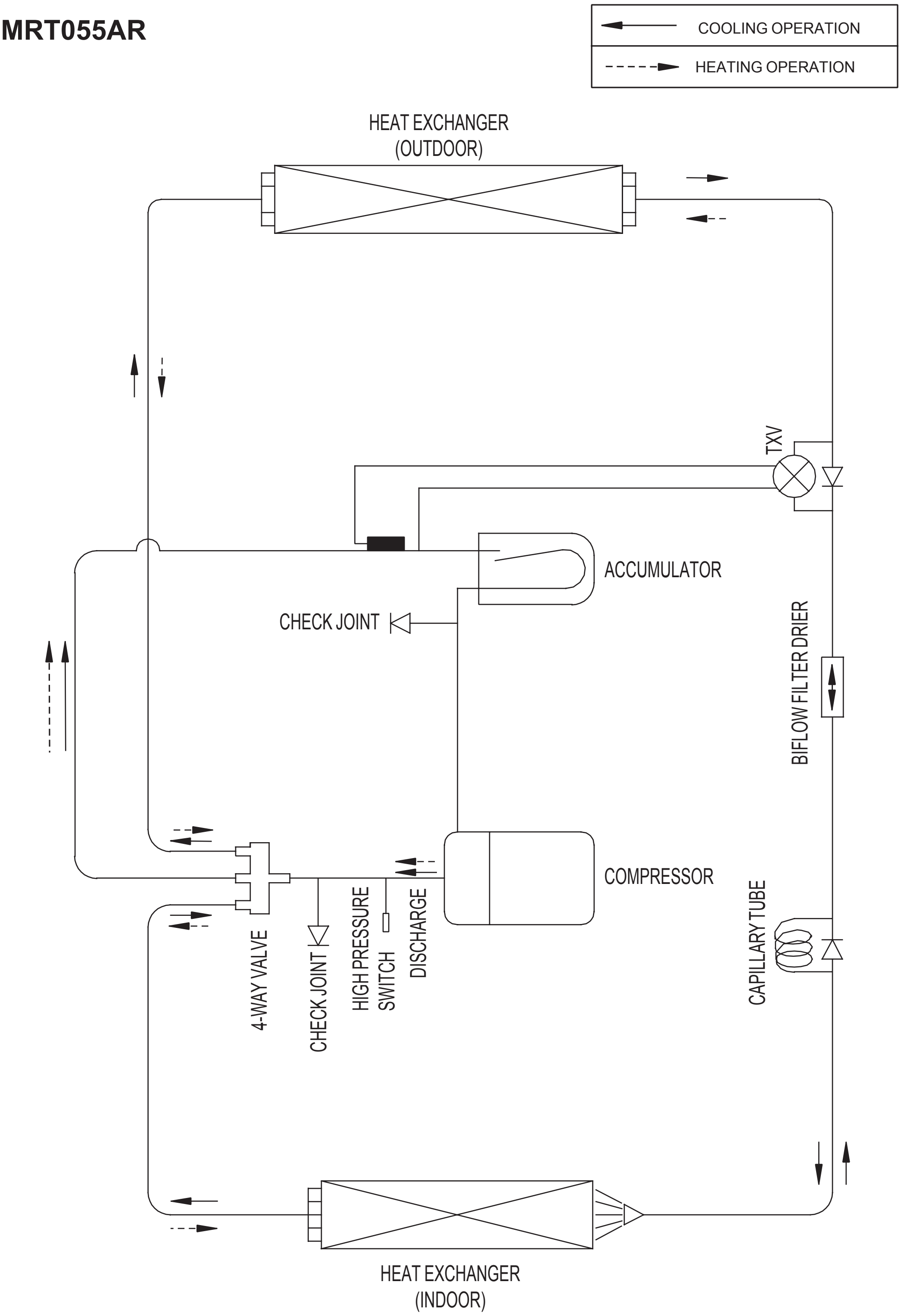
M(4)RT250/300A



M(4)RT360/420A

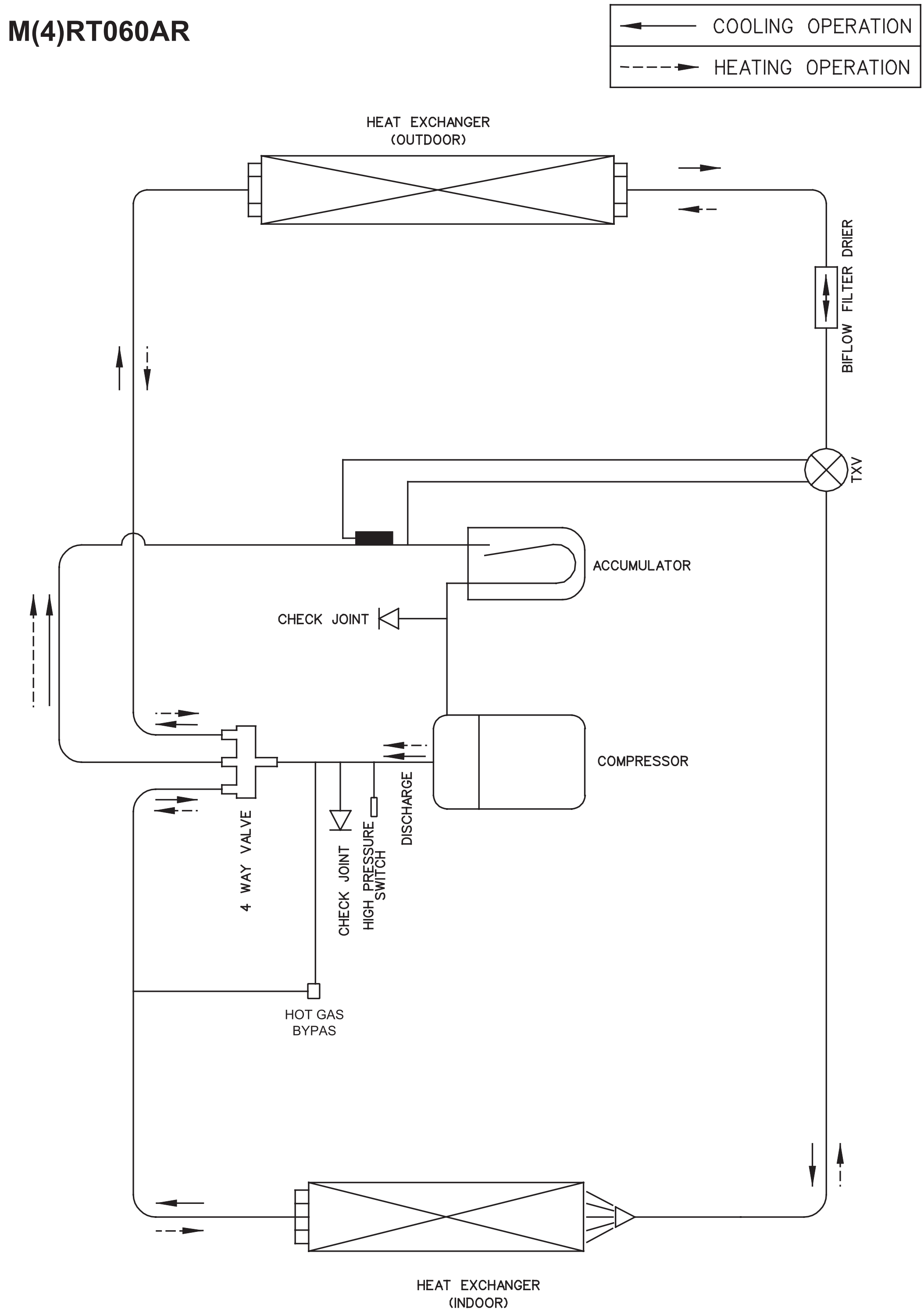


MRT055AR



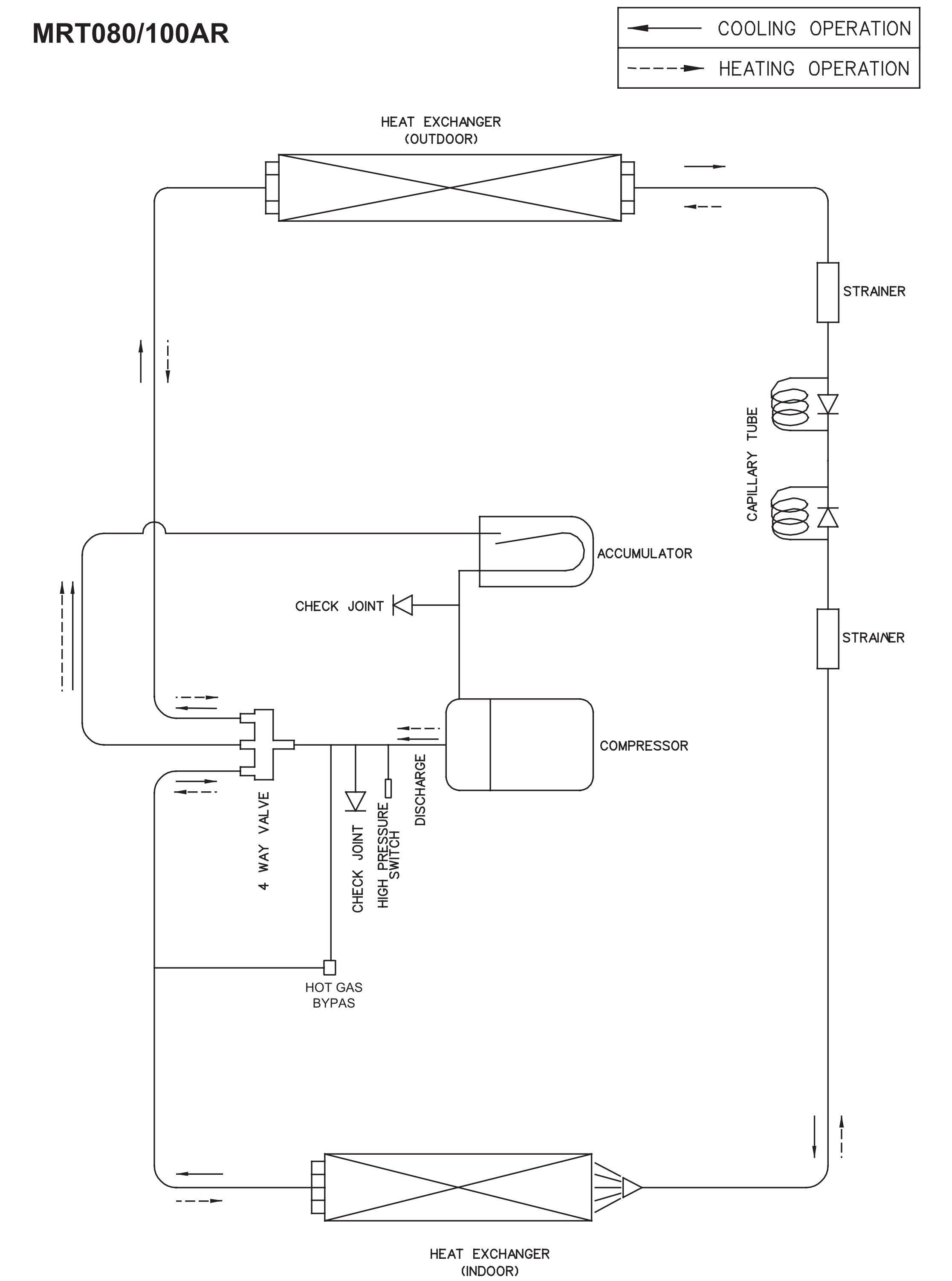
COOLING OPERATION
HEATING OPERATION
HEAT EXCHANGER
(OUTDOOR)
TXV
ACCUMULATOR
CHECK JOINT
BIFLOW FILTER DRIER
COMPRESSOR
CAPILLARY TUBE
4-WAY VALVE
CHECK JOINT
HIGH PRESSURE
SWITCH
DISCHARGE
HEAT EXCHANGER
(INDOOR)

M(4)RT060AR

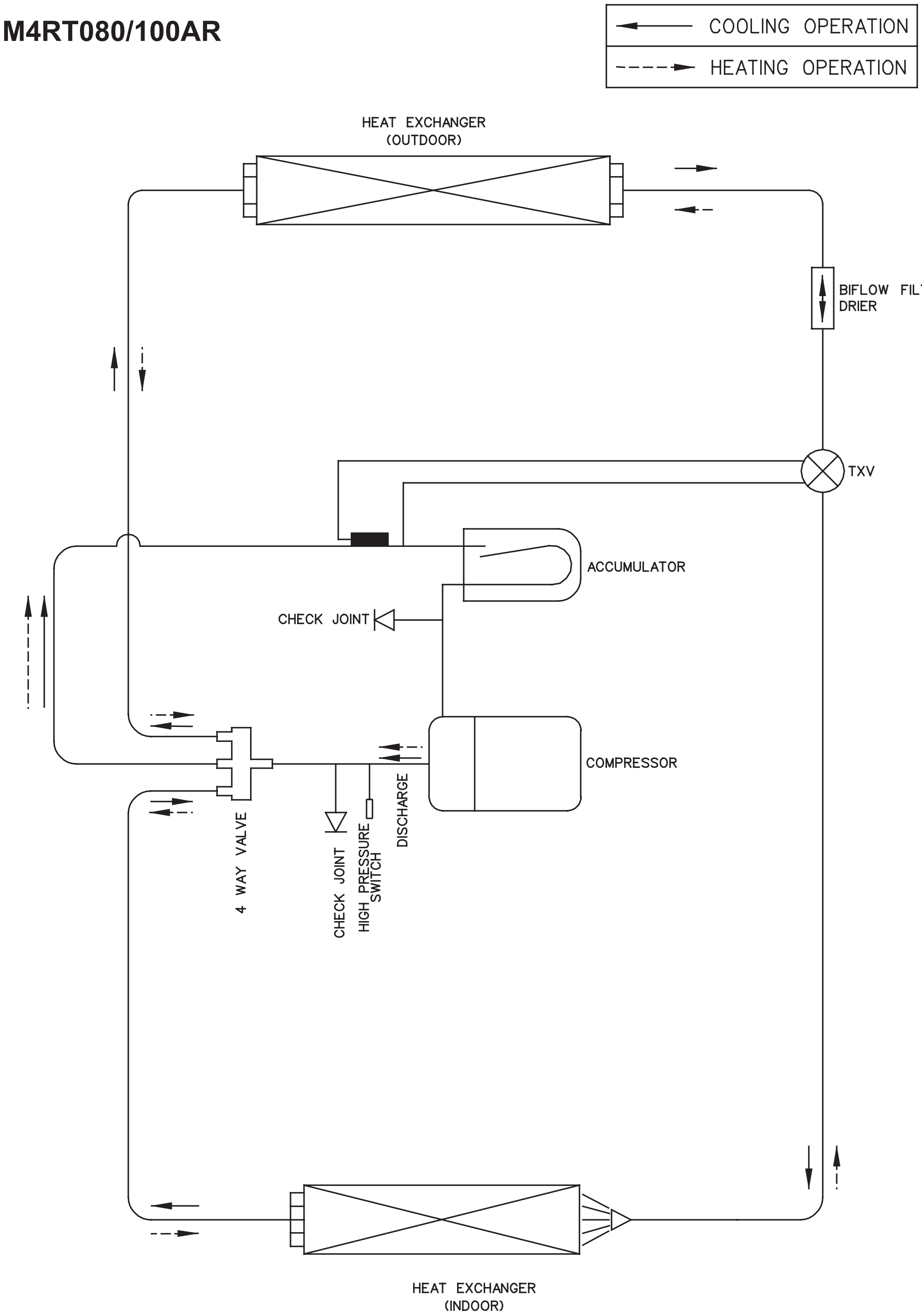


COOLING OPERATION
HEATING OPERATION
HEAT EXCHANGER
(OUTDOOR)
BIFLOW FILTER DRIER
TXV
ACCUMULATOR
CHECK JOINT
COMPRESSOR
DISCHARGE
HIGH PRESSURE
SWITCH
CHECK JOINT
4 WAY VALVE
HOT GAS
BYPAS
HEAT EXCHANGER
(INDOOR)

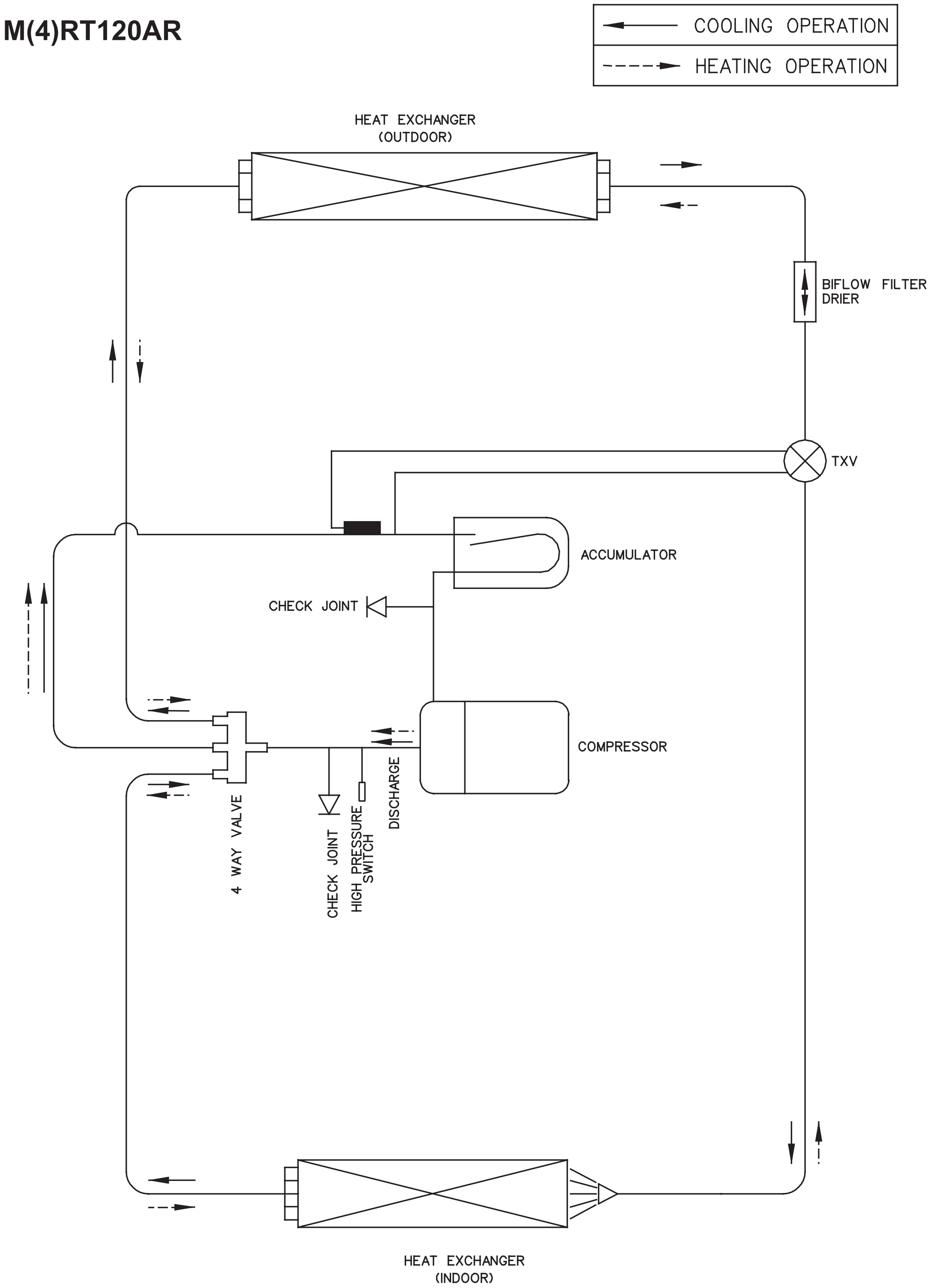
MRT080/100AR



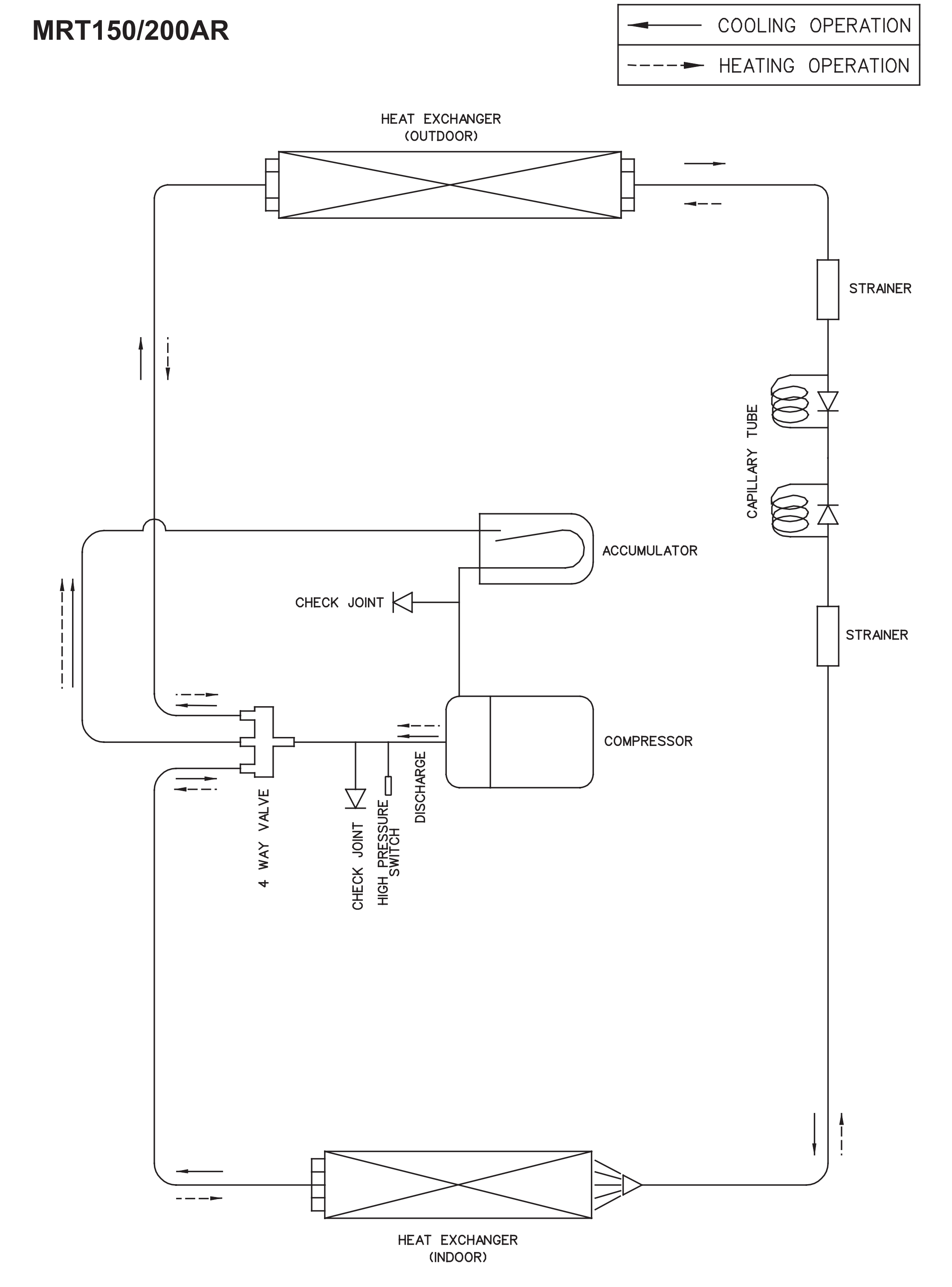
M4RT080/100AR



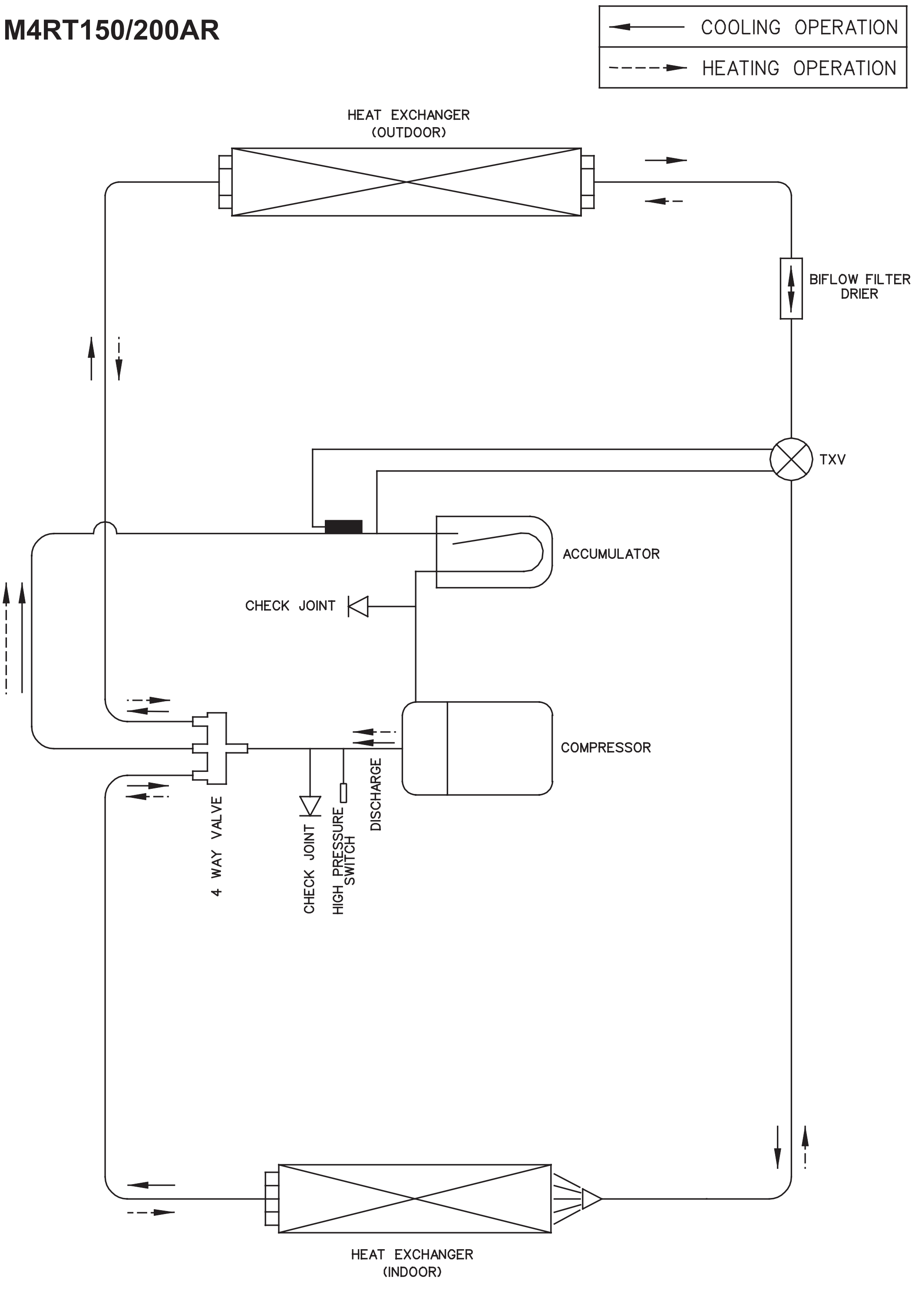
M(4)RT120AR


COOLING OPERATION
HEATING OPERATION
HEAT EXCHANGER
(OUTDOOR)
BIFLOW FILTER
DRIER
TXV
ACCUMULATOR
CHECK JOINT
COMPRESSOR
4 WAY VALVE
CHECK JOINT
HIGH PRESSURE
SWITCH
DISCHARGE
HEAT EXCHANGER
(INDOOR)

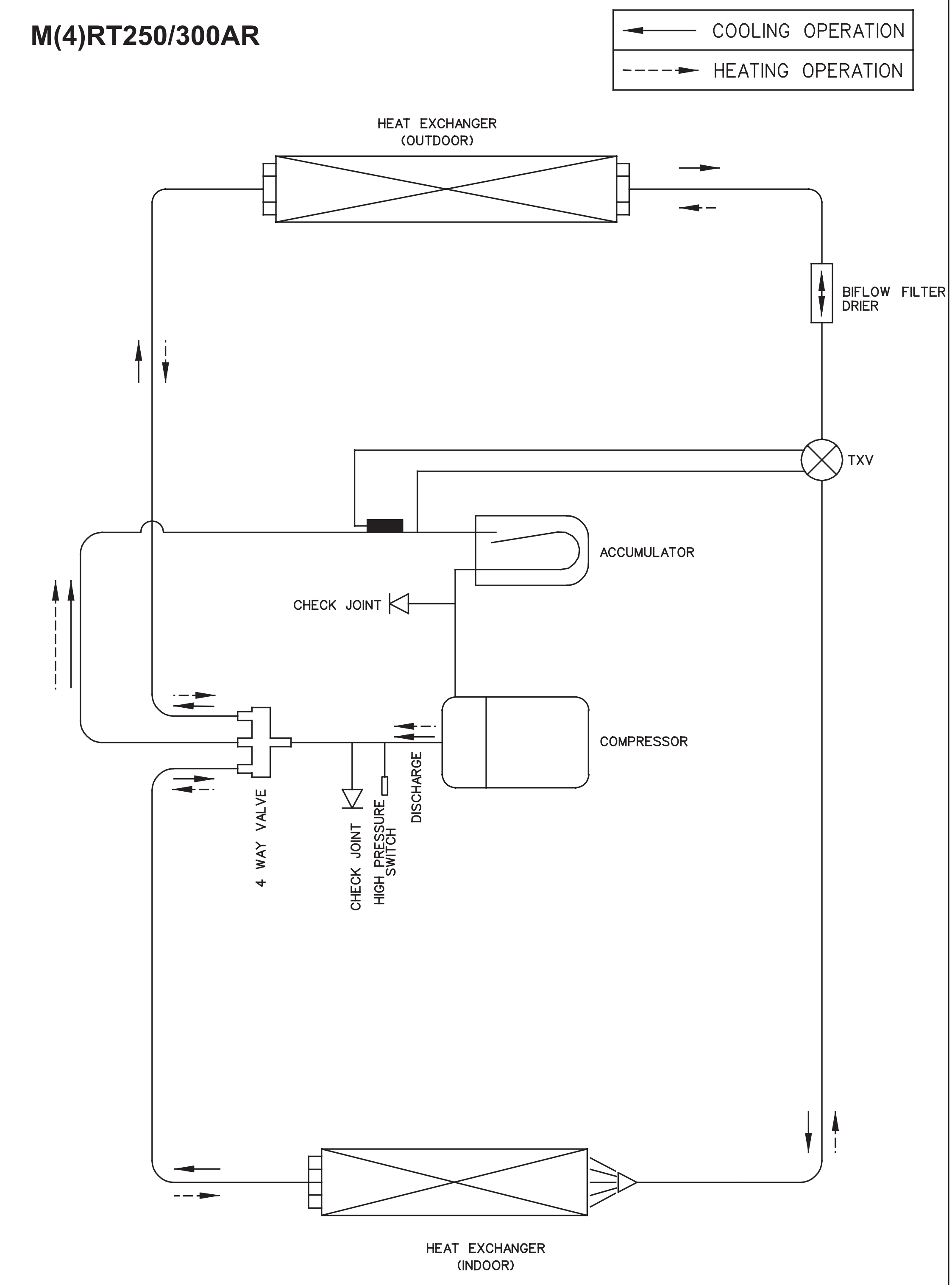
MRT150/200AR



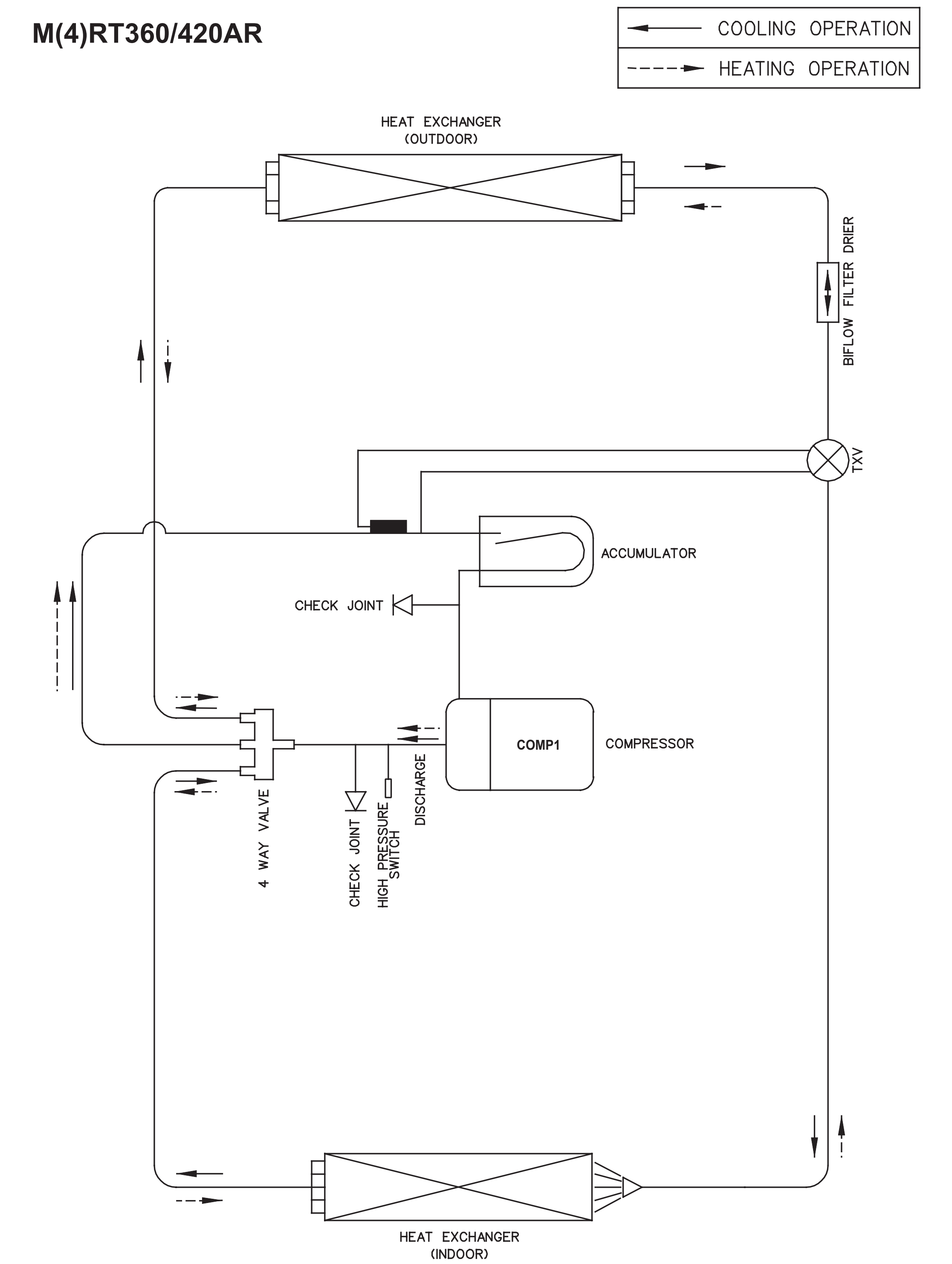
M4RT150/200AR



M(4)RT250/300AR

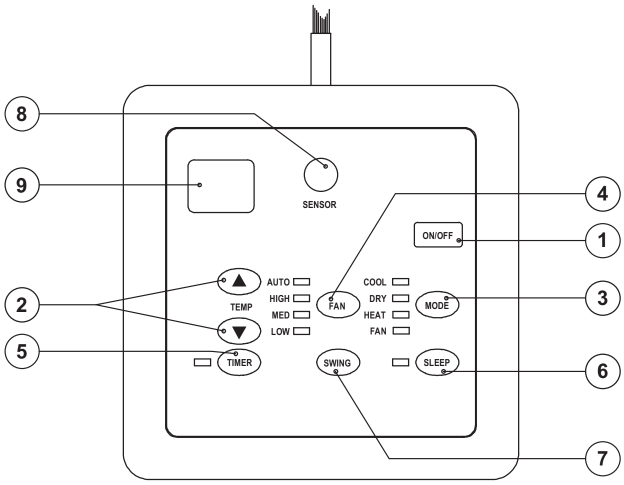


M(4)RT360/420AR

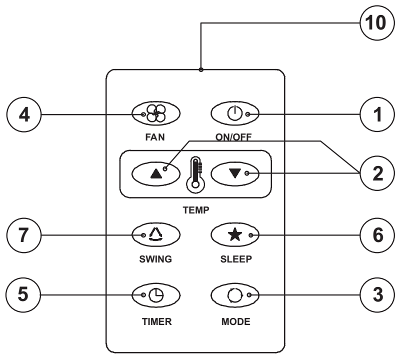


Controllers

SLM Wired Controller (Model: M(4)RT055/060/080/100/120A/AR)



SLM



AC-5300 (OPTIONAL)

1. **"ON/OFF" Switch**
   - Press to start the air conditioner unit.
   - Press again to stop the unit.

2. **Temperature Setting**
   - Set the desired room temperature.
   - Press button to increase or decrease the set temperature. Setting range are between16ºC to 30ºC (60ºF to 80ºF).

3. **Operation Modes**
   - Press the "mode" button for select the type of operating mode.
   - Cooling Only :
     COOL, DRY, FAN
   - Heat Pump :
     AUTO, COOL, DRY, HEAT, FAN
     (AUTO mode is represented by both COOL and HEAT LED light on)

4. **Fan Speed Selection**
   - Press the button until the desired fan speed is achieved.

5. **Timer**
   - Press the set button to select the switch timer of the air conditioner unit (the setting range is between 1 to 15 hours).

6. **"SLEEP" Mode**
   - Press button to activate the sleep function. This function can only be activated under "cool" or heating mode operation. When it is activated under ìcoolî mode operation, the set temperature will increase 0.5oC after 30 minutes, 1ºC after 1 hour and 2ºC after 2 hours. If it is activated under "HEAT" mode operation, the set temperature will be decreased 0.5ºC after 30 minutes, 1ºC after 1 hour and 2ºC after 2 hours.

7. **Air Swing**
   - Press button to activate the automatic air swing function.

8. **Sensor**
   - Infra red sensor to receive signals from wireless controller.

9. **LED Display**
   - To display the set temperature (inºC) and timer delay setting (in hours).

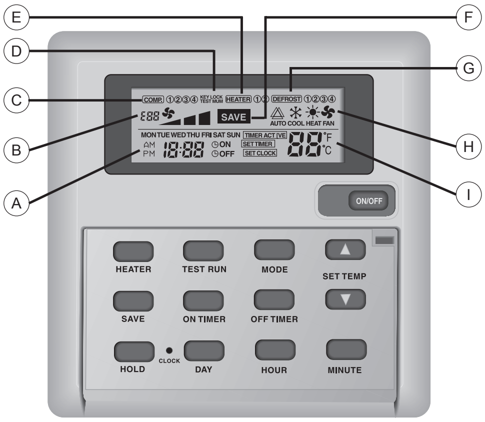
10. **Transmission Source**
    - To transmit signals to the air conditioner.

Error	Seven Segments
Room Sensor missing/Short	E1 blinking
Indoor coil sensor missing/Short	E2 blinking
Outdoor coil sensor missing/Short	E3 blinking
Compressor overload	E4 blinking
Outdoor abnormal compressor overload trip or gas leak	E5 blinking

Sequential Controller (Model: M(4)RT150/200/250/300/360/420A/AR)

Sequential Controller Functions

1) Sequential Controller LCD Display



A : Time display
B : Error indication
C : Compressor running display (up to 4 compressors)
D : Key lock display
E : Heater display (up to 2 heaters)
F : Energy saving mode display
G : Compressor defrost cycle display (up to 4 compressors)
H : Operation mode display
I : Temperature set display

2) Operating Guide

a) ON/OFF Key

Press once to start the air conditioning unit.
Press again to stop the unit.
The operation lamp next to the key lights up and goes off respectively when the unit is running or not running.
Caution : In the case when the ON/OFF key is pressed immediately after the operation is stopped, the unit will not restart until 3 minutes later to protect the compressor.

b) Selecting Operation Mode

Press the MODE key to select the type of operating mode. Consecutive press of the key switches the operation over “COOL”, “HEAT”, “AUTO” and “FAN”

c) Save Mode

Press the SAVE key to select the energy saving function. This option is only available for “COOL”, “HEAT” and “AUTO” modes.

d) Auxiliary Electric Heater

If the “HEAT” mode provides insufficient heating to a room even at the highest temperature setting (30ºC), press the HEATER key to activate the auxiliary electric heater. For models with two heaters, consecutive press of the key allows the selection of one or both heaters active.

e) Temperature Setting

To set the desired room temperature, press ▲ or ▼ to increase or decrease the set temperature in the range of 16ºC to 30ºC. Press both ▲ and ▼ simultaneously to toggle between ºC and ºF setting.

f) Time Setting

Real time Clock

Press the CLOCK key once to activate set clock mode.
Press again to disable set clock mode.
Under set clock mode, the time of the present day can be set by pressing the respective MINUTE, HOUR and DAY key.

7-days timer
Press the **ON TIMER** key to activate auto-ON timer mode. Under this mode, press the respective **MINUTE, HOUR** and **DAY** key to select the time of the week when the air-conditioning unit is to automatically start running. Press the **ON TIMER** key again to save the setting.

Press the **OFF TIMER** key to activate auto-OFF timer mode. Under this mode, press the respective **MINUTE, HOUR** and **DAY** key to select the time of the week when the air-conditioning unit is to automatically stop running. Press the **ON TIMER** key again to save the setting.

Then to activate the 7-days timer, press and hold the **TIMER ACTIVE** key until the word "TIMER ACTIVE" appears on the LCD screen. Repeat the same step to disable the 7-days timer.

g) Other Function

Key Lock
Press the **MINUTE** key 3 times consecutively and fast to activate the key lock. A "KEY LOCK" symbol will appear on the LCD screen. At this point, only the ON/OFF key is valid.
To disable the key lock, again press the **MINUTE** key 3 times consecutively and fast.

Test run
Press the TEST key 2 times consecutively to test run the unit.

3) Error Code

When the system is on and an error occurs, the **ON/OFF** LED on the LCD panel will blink and an error code is shown. When the system is off and there is a thermistor error, the **ON/OFF** LED is off but the error code is still displayed. Each error code represents different message as below

Error code	Possible fault	Error code	Possible fault
E01	Require manual reset (possible causes)		
E02	Compressor 1 high temperature (overload)	E20	Indoor coil sensor 1 open
E03	Compressor 2 high temperature(overload)	E21	Indoor coil sensor 2 open
E06	Compressor 1 high pressure trip / contact open	E24	Outdoor coil sensor 1 short
E07	Compressor 2 high pressure trip / contact open	E25	Outdoor coil sensor 2 short
E10	Compressor 1 trip / low R-22 / outdoor abnormal	E28	Outdoor coil sensor 1 open
E11	Compressor 2 trip / low R-22 / outdoor abnormal	E29	Outdoor coil sensor 2 open
E14	Room sensor short	E32	Compressor 1 de-ice
E15	Room sensor open	E33	Compressor 2 de-ice
E16	Indoor coil sensor 1 short		
E17	Indoor coil sensor 2 short		

4) Installation Of LCD Remote Controller

a) Accessories

The following accessories are included. If any part is missing, contact your dealer immediately.
① Remote controller
② Wooden screw 4.1 x 16 (2 pieces)
③ Instruction manual

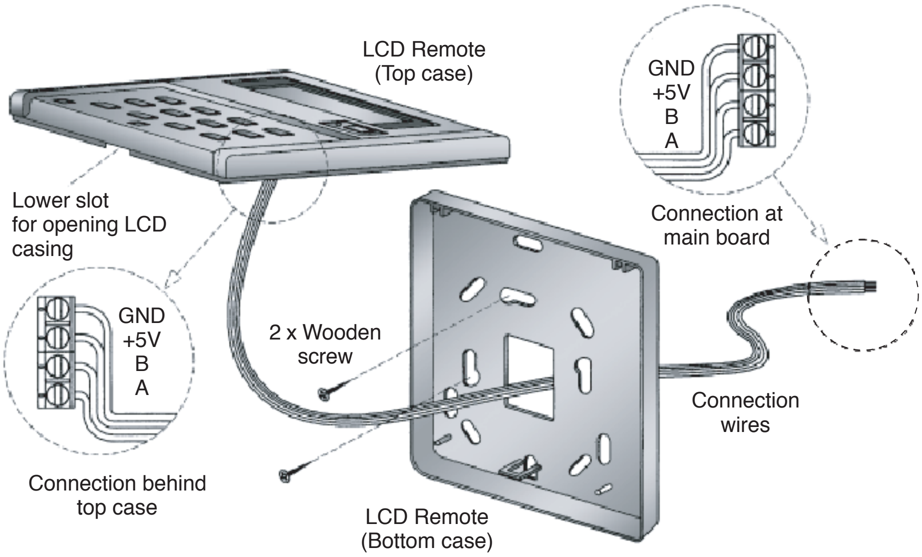
b) Step-by-step guide

i) First, open up the casing of the LCD remote controller into its top and bottom case using a screwdriver. To do this, insert the screwdriver into the lower slot and slide it in the outward direction.

ii) Fix the bottom case onto the wall with the 2 wooden screws provided. Then, insert the 4 connecting wires (from the main board) through the slot on the lower right.

iii) Connect one end in each of the 4 wires to the terminal block behind the top case as shown below. The wire that goes into the “GND” terminal at the top case must be connected at the other end to the “GND” terminal at the main board. The same goes for the “+5V”, “B” and “A” connection.

iv) Fasten back the top and bottom case into place. Hook the two upper claws into their respective slots and snap the lower part shut.



5) Auto Random Restart

When power is resumed, the unit will automatically restart and operate at the previous setting as before power failure occurred. (Remove jumper at JH/JP1 to cancel the auto random restart function. Please refer to wiring diagram for the location of JH/JP1 Jumper).